PROJECT DOCUMENTATION

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

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PURATCHI THALAIVER DR. MGR GOVERNMENT ARTS & SCIENCE COLLEGE

College Code: TVU364

B.Sc. Computer Science

GROUP PROJECT

Academic Year 2026–2027

PROJECT TEAM DETAILS

The project is prepared as a group project. The following member information is presented individually for clear identification in the final submission.

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1. PROJECT OVERVIEW

The Student Enrollment & Course Management System is an academic management project intended to organize student information, course information and enrollment activities in a single structured application. The project focuses on making academic records easier to create, maintain, search and review. Instead of keeping student and course details as separate, disconnected records, the system connects them through an enrollment process. This creates a clear relationship between a student and the course selected by that student.

The project interface references supplied by the team show dashboard-style screens, student-management views and course-management views. These visual references are used in this documentation only as project evidence. The actual project concept is organized around the three core data areas: Student, Course and Enrollment. Supporting features such as validation, reports, dashboard summaries, user access and testing are included to make the application workflow complete.

The Student area stores the identity and academic details of learners. The Course area maintains the available courses and their descriptive information. The Enrollment area records the registration relationship between a particular student and a particular course. This relationship is important because it allows the system to answer practical questions such as which courses a student has selected, how many students are registered for a course, and which enrollments are active.

The project is designed with a simple professional user flow. A user first opens the application and verifies the required data. Student and course records are created before an enrollment is entered. During enrollment, the user selects the existing student and course records, enters the required registration information and saves the transaction. After saving, the relationship is verified through related information, reports and dashboard views.

The project also demonstrates the importance of data quality. Required fields should be completed before a record is saved. Relationship fields should refer to valid Student and Course records. Status and date values should follow the rules configured for the application. These checks reduce incomplete or inconsistent records and make later reporting more reliable.

Overall, the project demonstrates how a cloud-based academic application can combine data management, relationships, validation, reporting and dashboard presentation in one workflow. The following sections explain how the project is organized and how the complete process can be demonstrated step by step.

2. PROBLEM STATEMENT AND OBJECTIVES

Managing student enrollment manually can require repeated entry of the same information and may make it difficult to understand the relationship between students and courses. A student may appear in one record, a course may appear in another record and the registration relationship may be maintained separately. When the number of records increases, searching and updating this information becomes more time-consuming. The project addresses this problem by organizing the information into connected records.

The first objective is to maintain student details in a structured Student record. The record can contain identifying information, contact information, department, year and status. The second objective is to maintain a Course catalogue containing course code, course name, credits, department, description and status. The third objective is to create an Enrollment record that connects a selected student with a selected course and stores the registration information.

Another objective is to reduce data-entry errors. Required fields and validation rules can prevent incomplete records. A user should be able to understand what information is required and receive a clear message when an entry does not satisfy the configured rules. This improves consistency before the information reaches reports and dashboards.

The project also aims to provide quick access to information. A user should be able to search and open a student record, view course information, review enrollment details and generate reports without manually checking multiple unrelated files. Dashboard components can summarize the available information so that the project demonstration can show both individual records and overall statistics.

The system is also designed with controlled access in mind. Different users may require different permissions. Administrative users may manage records and configuration, while other users may only need to view or update selected information. Access control should therefore be tested as part of the project rather than treated as a separate activity.

The main objectives can be summarized as follows:

- Centralize student, course and enrollment information.
- Connect Student and Course records through Enrollment.
- Improve data consistency through validation.
- Provide reports and dashboards for quick review.
- Support controlled access according to user responsibility.

3. SCOPE AND SYSTEM REQUIREMENTS

The scope of the project covers the core activities required to manage student enrollment and courses. The Student Management module handles creation, viewing and updating of student records. The Course Management module maintains the available course catalogue. The Enrollment Management module connects students and courses and records registration information. Reports and dashboards provide summarized views of the stored data.

The system requires a user interface through which authorized users can navigate to the required modules. It requires a data model capable of storing Student, Course and Enrollment information and maintaining the relationships among those records. It also requires validation so that important information is not saved in an incomplete or invalid form. Reporting and dashboard components are required for presenting the stored information in a useful format.

Functional requirements include creating student records, creating course records, creating enrollment records, viewing existing records, editing records when permitted, searching records, validating required information, producing reports and displaying dashboard summaries. The system should also allow the project team to demonstrate the relationship between Student, Course and Enrollment data.

Non-functional requirements include usability, consistency, maintainability, security and reliability. The interface should use clear labels and logical navigation. Records should be presented in a readable manner. The system should preserve data relationships when records are created or updated. Users should receive meaningful validation messages when an action cannot be completed.

The project can be implemented on a Salesforce environment, using standard platform capabilities for data objects, relationships, permissions, reports and dashboards. Where custom behaviour is required, suitable Salesforce development components can be used according to the project configuration. The exact features available in the final environment depend on what has been configured and enabled by the project team.

The basic scope does not require complete fee payment processing, biometric attendance, full examination management, hostel administration or external ERP integration. Those features can be considered later if the project is expanded. Keeping the current scope focused allows the team to demonstrate the complete student-to-enrollment workflow clearly.

The requirements therefore support a complete academic demonstration: create or verify master records, perform an enrollment transaction, validate the input, confirm the relationship, generate reports and view dashboard information.

- Student Management
- Course Management
- Enrollment Management
- Validation and data-quality controls
- Reports and dashboards
- User access and security

4. DATA MODEL AND MODULES

The data model is centered on three major entities: Student, Course and Enrollment. The Student entity represents a learner. It contains information required to identify and manage the learner, such as Student ID, Student Name, Email, Phone, Department, Year and Status. A Student record acts as one of the master records used during enrollment.

The Course entity represents an academic course. Important information can include Course Code, Course Name, Description, Credits, Department and Course Status. The Course record forms the second master record used during enrollment. Maintaining course information separately helps keep course selection consistent.

The Enrollment entity represents the relationship between a student and a course. It can contain Enrollment ID, Student, Course, Enrollment Date, Enrollment Status and Remarks. The Student and Course fields connect the enrollment to the corresponding master records. A student can have more than one enrollment when multiple courses are selected, and a course can have multiple students enrolled in it.

The Student Management Module provides the interface for creating and maintaining student records. The user enters the required details, saves the record and later searches or opens it when needed. The module provides the data that will be selected during enrollment.

The Course Management Module provides the course catalogue. Users can create a course, enter its identifying details and maintain its status. Course information can then be reused during enrollment and reporting rather than being typed repeatedly.

The Enrollment Management Module is the central transaction module. The user selects an existing Student record and an existing Course record, enters the enrollment date and status and saves the transaction. The saved record establishes the relationship between the two master records.

The Reporting and Dashboard Module provides a summary of the stored information. Reports can show student lists, course lists and enrollment lists. Dashboard components can show counts, distributions or other summaries based on the configured reports.

The modules are therefore connected rather than isolated. Student and Course provide the master information; Enrollment creates the relationship; Reports and Dashboards present the resulting data. This structure makes the project easier to explain, test and demonstrate.

- Student → learner master record
- Course → course master record
- Enrollment → Student–Course relationship
- Reports → detailed and summarized information
- Dashboard → visual project overview

5. SYSTEM WORKFLOW – STEP BY STEP

Step 1 – Login and open the application. The authorized user signs in and opens the configured Student Enrollment & Course Management application. The home or dashboard screen provides navigation to the required areas.

Step 2 – Create or verify Student information. Open the Student Management area and select the option to create a student. Enter the required student details such as student ID, name, email, department, year and status. Check the values before saving. After saving, search for the student and verify that the record is available.

Step 3 – Create or verify Course information. Open the Course Management area and create the required course. Enter the course code, course name, credits, department, description and status. Save the record and verify that it appears in the course catalogue.

Step 4 – Start the Enrollment transaction. Open the Enrollment area and select the option to create a new enrollment. The form should provide relationship fields for Student and Course. Select the correct existing records instead of manually entering unrelated text.

Step 5 – Enter enrollment details. Provide the enrollment date, enrollment status and any other required fields configured for the project. Before saving, check that the selected student and course are correct.

Step 6 – Apply validation. The system checks required fields and configured rules. If information is missing or invalid, the user corrects the entry and tries again. A valid record is then saved successfully.

Step 7 – Verify the relationship. Open the saved Enrollment record and confirm that it points to the intended Student and Course. Open the related Student or Course view where available and check that the enrollment appears correctly.

Step 8 – Generate reports. Run the configured Student, Course and Enrollment reports. Use filters or grouping when required. Confirm that the newly created records are included in the expected report.

Step 9 – Review the dashboard. Open the dashboard and check the summary components. The dashboard should present the configured totals or visual summaries based on the underlying reports and records.

Step 10 – Perform final testing and demonstration. Repeat the main workflow with valid data, demonstrate one validation scenario, verify the relationship and show the report and dashboard results. This creates a complete demonstration from data creation to final output.

6. VALIDATION, SECURITY AND DATA QUALITY

Validation is used to ensure that important information is entered before a record is stored. In the Student module, required identity fields should not be left blank. In the Course module, the course code and course name should be available before the course is saved. In the Enrollment module, both the Student and Course relationship fields should be selected so that the registration cannot exist without its required connections.

Validation should be tested with both valid and invalid input. For a valid case, the user completes all required information and the system saves the record. For an invalid case, the user deliberately leaves a required field empty or enters a value that violates a configured rule. The system should prevent the save and display a message that helps the user correct the problem.

Data quality also depends on using existing relationship records correctly. During enrollment, selecting an existing Student and Course is preferable to entering inconsistent names manually. The relationship ensures that the enrollment is connected to the correct master data and can later be used in reports.

Security is another important part of the project. Access should be provided according to user responsibility. Profiles and permission sets can control access to objects and fields, while sharing configuration can control record visibility where required. The team should test that an authorized user can perform the required actions and that a restricted user cannot perform actions outside the configured permissions.

Data quality should be checked after updates as well. If a student changes status or a course becomes inactive, the effect on future enrollment activity should follow the rules configured for the project. The exact business rules depend on the final configuration, so the demonstration should use the rules actually implemented.

A good project demonstration therefore includes three checks: first, valid data should be accepted; second, invalid or incomplete data should be controlled; and third, users should only receive the access required for their role. These checks show that the project is not only a collection of screens but a structured information-management system.

7. REPORTS, DASHBOARD AND USER INTERFACE

Reports provide organized lists and summaries of the records stored by the application. A Student Report can display student information, a Course Report can display available courses, and an Enrollment Report can display registration details. Filters can be used to focus on active records, a particular department, a particular course or another configured category.

The dashboard presents selected report information visually. Typical components can include total students, active students, total courses, active courses and enrollment-related summaries. The supplied project references include dashboard-style interfaces with cards, charts and quick-access areas. These images are used in the final screenshot section as visual references supplied by the team.

The user interface should make the workflow easy to follow. Navigation should clearly identify the main modules. A user should be able to move from Student Management to Course Management and then to Enrollment without confusion. Actions such as New, Save, Edit and View should be clearly presented.

A professional interface should also keep the information hierarchy clear. The page title should identify the current area, important summary information should appear before detailed records, and related actions should be grouped logically. Search and filtering controls should be easy to locate when the record list is large.

The dashboard is particularly useful during project presentation because it provides a quick visual summary. However, dashboard values should be understood as results of the underlying records and reports. The team should therefore verify the records before presenting the dashboard.

For demonstration, the recommended sequence is to show the dashboard briefly, open Student Management, show a student record, open Course Management, show a course record, create or open an Enrollment record, and then return to the reports and dashboard. This sequence connects the individual data records with the final summary.

The supplied screenshots also show different reference interfaces such as course management tables, student management dashboards, administrative dashboards and school dashboards. Only the six requested project-image areas are included in the final documentation, and they are cropped from the supplied screenshots so the actual interface is visible at a useful size.

8. TESTING AND PROJECT DEMONSTRATION

Testing verifies that each part of the project works correctly and that the modules work together. The first level is module testing. Create a Student record and confirm that it can be saved and searched. Create a Course record and confirm that it appears in the course catalogue. Create an Enrollment record and confirm that it connects the correct Student and Course.

The second level is relationship testing. After an enrollment is created, open the enrollment and verify the selected Student and Course. Then open the related record where the configured relationship allows it and check whether the enrollment is displayed correctly. This confirms that the data model is functioning as intended.

The third level is validation testing. Attempt to save a Student without a required field, a Course without a required field and an Enrollment without one of its required relationships. The system should respond according to the validation rules configured by the project team.

The fourth level is reporting testing. After test records are created, run the configured reports and confirm that the expected records appear. Check filters and grouping if they are used. Then open the dashboard and verify that the displayed information corresponds to the underlying report data.

The fifth level is security testing. Use the available user roles or permission settings and verify that each user can perform the actions assigned to that role. Any restricted action should remain protected.

For the final project demonstration, the team can follow this order: open the application, show the dashboard, create or open a Student record, create or open a Course record, create an Enrollment record, demonstrate validation, verify the relationship, show the reports, and finally show the dashboard again. This sequence explains the complete project in a logical order.

The final presentation should focus on what is actually configured in the project. If a feature has not been implemented, it should not be presented as an implemented feature. The screenshots at the end are visual evidence of the supplied project references and are separate from the written workflow.

9. PROJECT MODULE SUMMARY AND OPERATIONAL FLOW

The complete operational flow can be understood as a chain of related activities. Student Management creates the learner record. Course Management creates the academic course record. Enrollment Management combines those two records into a registration transaction. Validation protects the quality of the information. Reports organize the records for review. Dashboards summarize the resulting information for quick understanding.

Student Management begins with the creation of a unique student record. The user enters the required information and saves it. Once the student exists, it becomes available for selection during enrollment. Course Management follows the same master-data principle: a course is created once and then reused when different students enroll.

Enrollment is therefore the connecting point of the system. The user does not create an isolated registration. Instead, the user selects the existing Student and Course records and creates a relationship between them. The enrollment date and status describe the transaction.

After the enrollment is stored, the project team verifies the data. The verification can be performed by opening the enrollment record, checking the selected relationships and confirming that the record appears in the appropriate list or report. If the configured interface provides related lists, those can also be used for verification.

The reporting stage transforms the stored records into readable information. A list report can display detailed rows, while a summary report can group records according to a selected field. Dashboard components then provide a visual overview based on the configured report data.

The final operational flow is therefore:

User Login → Dashboard/Application Home → Student Management → Course Management → Enrollment Management → Validation → Relationship Verification → Reports → Dashboard Review → Final Testing.

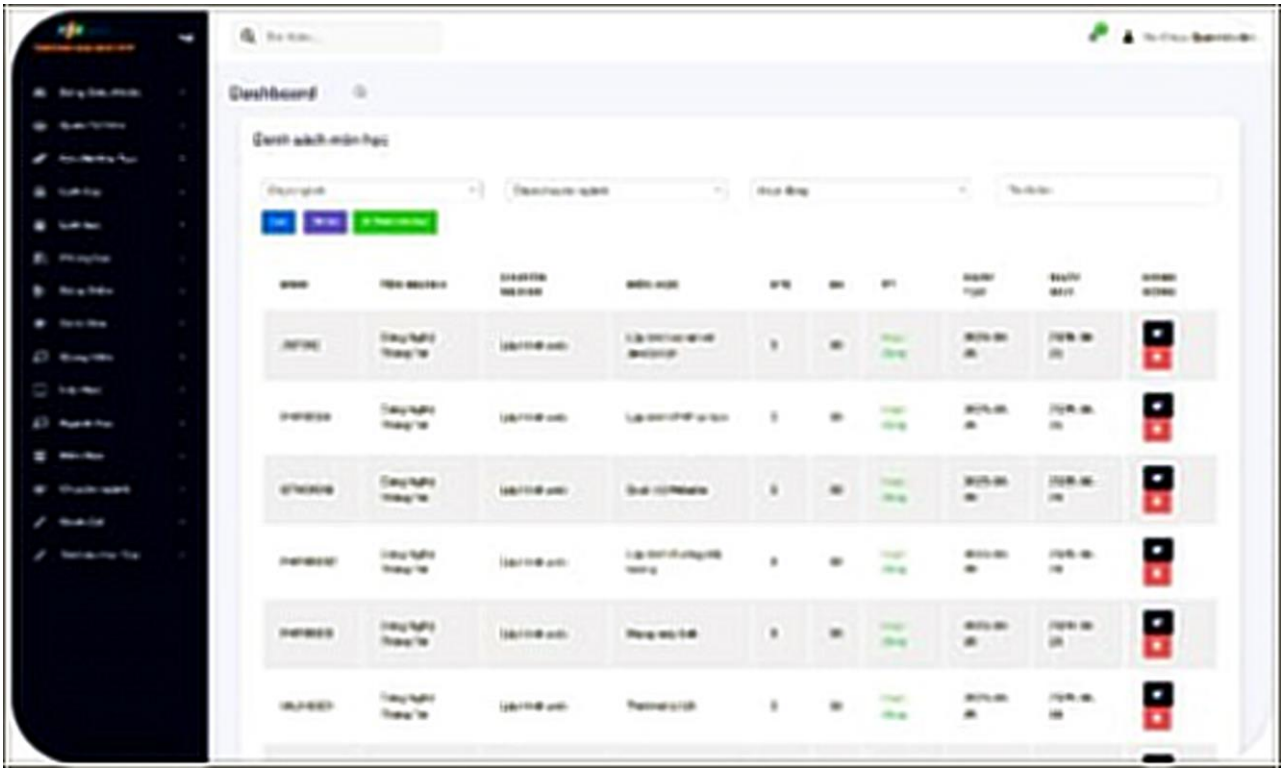
For project presentation, each step should be shown with the corresponding screen. The team can explain what data is entered, what the system checks, what record is created and how that record becomes part of the final report or dashboard. This makes the project workflow understandable even to a reviewer who has not used the application before.

This step-by-step structure is the main demonstration path of the project and should be followed consistently during the final presentation and submission review.

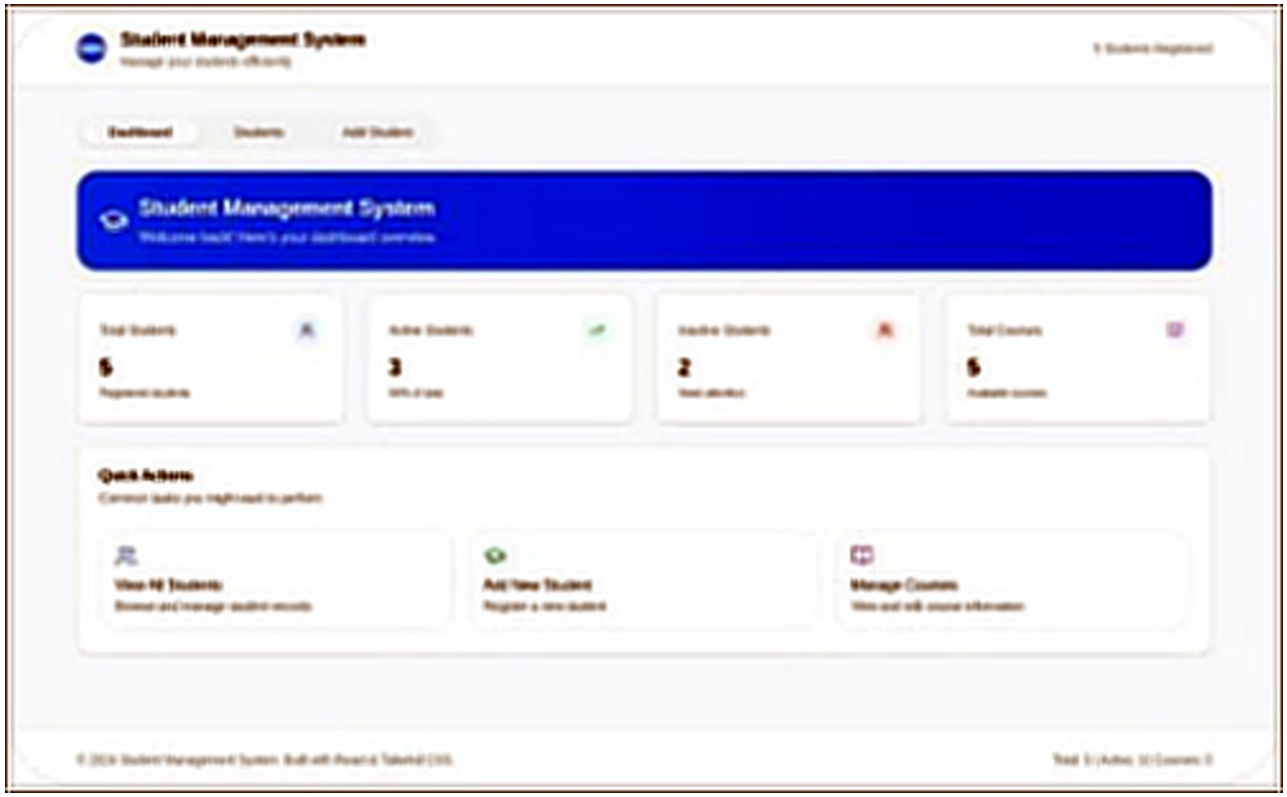
10. PROJECT SCREENSHOTS

Only the six requested screenshots are included below. The surrounding phone/chat content has been removed; each page contains the small project interface visible in the middle of the supplied screenshot, enlarged proportionally for clear viewing.

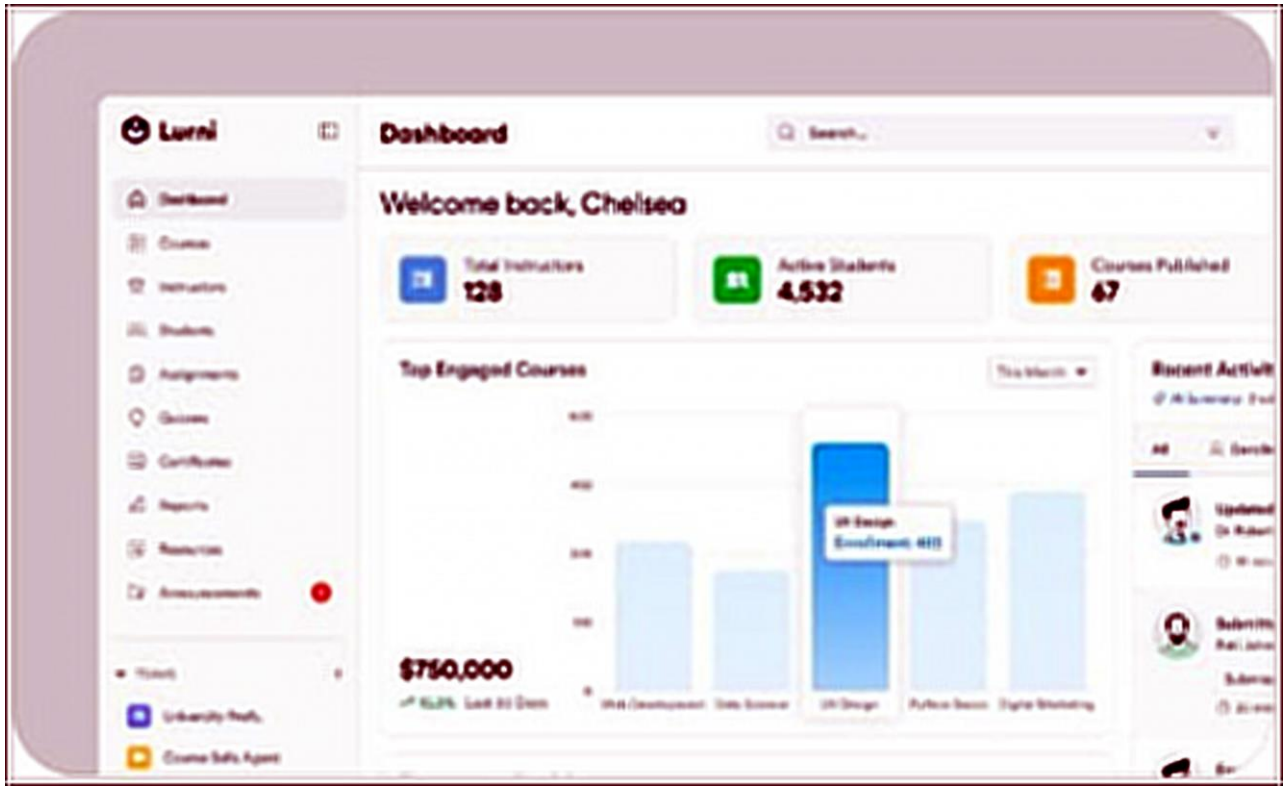
PROJECT SCREENSHOT 1



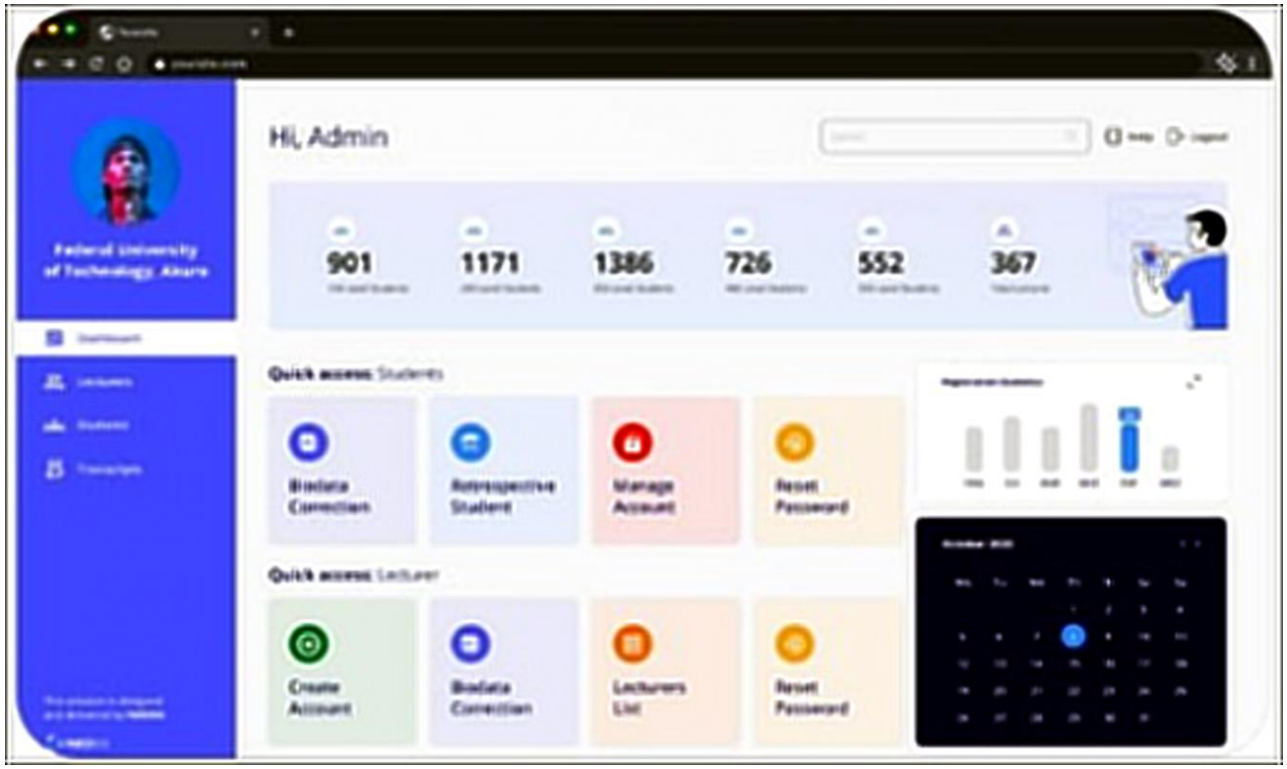
PROJECT SCREENSHOT 2



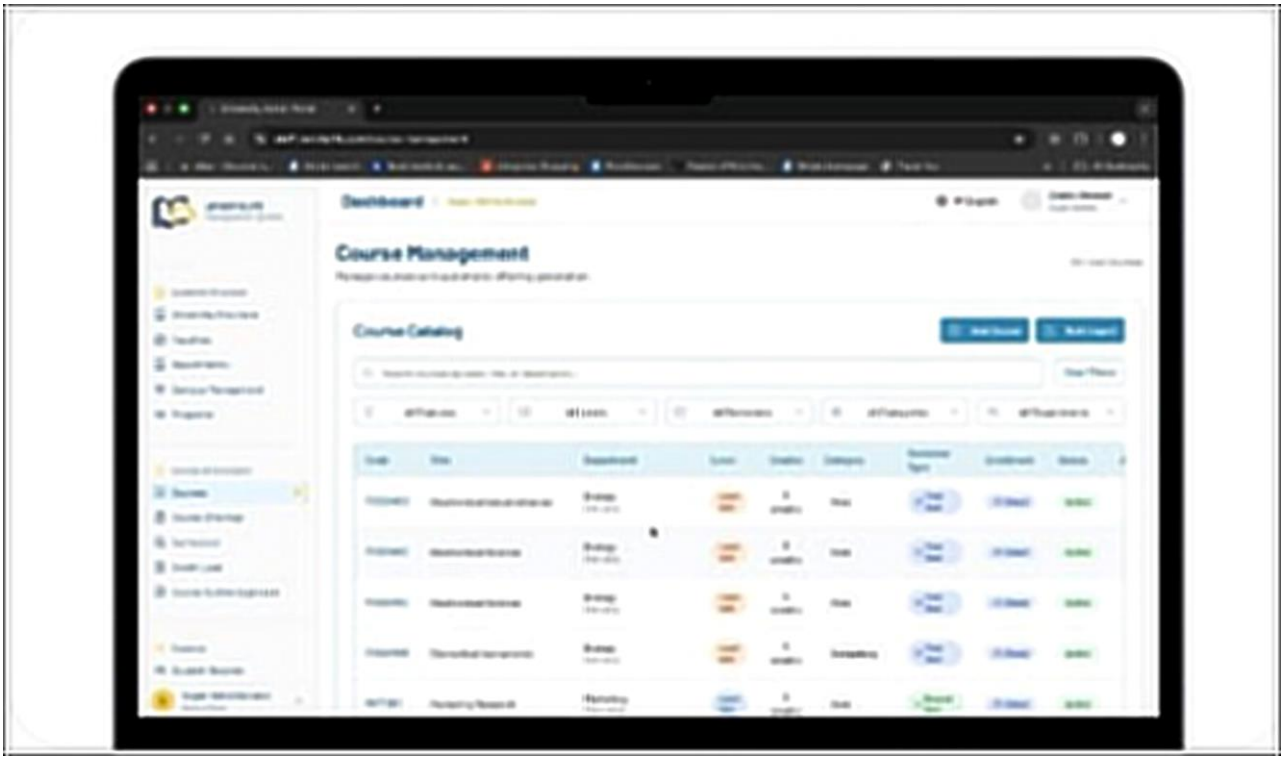
PROJECT SCREENSHOT 3



PROJECT SCREENSHOT 4



PROJECT SCREENSHOT 5



PROJECT SCREENSHOT 6



11. CONCLUSION

The Student Enrollment & Course Management System provides a structured approach to managing student information, course information and enrollment activities in one connected application. The documentation presents the project as a complete operational flow rather than as separate screens.

The project begins with student and course master information. Student records provide the learner details required for academic management, while course records provide the course code, name, credits, department, description and status. The Enrollment module connects these records and records the actual student–course relationship.

Validation and data-quality checks are important throughout the workflow. Required information should be entered before saving a record, and enrollment should use valid existing Student and Course records. The workflow also includes relationship verification, reporting and dashboard review so that the team can demonstrate not only data entry but also how the stored information is reviewed.

The supplied project screens demonstrate course management, student management, dashboards, administration and school-level summary views. Together with the step-by-step workflow, these screens provide a practical basis for explaining the system during project demonstration and submission review.

The project therefore demonstrates a clear academic management process: user access, student management, course management, enrollment, validation, relationship verification, reporting and dashboard presentation. The final documentation is organized so that each topic and its supporting explanation can be reviewed in sequence.

12. REFERENCES

1. Salesforce Trailhead – learning resources for Salesforce platform, data management and application configuration.
2. Salesforce Help Documentation – platform concepts, objects, fields, relationships, permissions, reports and dashboards.
3. Salesforce Platform / Lightning Experience – reference for the interface patterns represented in the project screens.
4. Project screenshots supplied by the project team – used as the visual reference for the documented system screens.
5. Project team notes and workflow description – used to organize the Student, Course, Enrollment, Validation, Reporting and Dashboard flow.

Note: The references above identify the platform and project materials used for documentation. The screenshots are reproduced from the images supplied for this project.

THANK YOU

Student Enrollment & Course Management System

Project Documentation – Final Submission

College Code: TVU364

Thank you for reviewing our project documentation.